

LESSON 302 (1.0 AATD)

Lesson Objective

During this lesson, the student is provided with an introduction to attitude instrument flying with emphasis on learning aircraft control by reference to instruments.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)

Introduction to the Simulator

- Discuss maneuvers, procedures, and introduce the instrument cockpit check.
- Highlight the sensitivity of the FRASCA MENTOR and emphasize the need for small control inputs.
- System power up:
 - Student Tells, Student Does.
 - Have the student start the simulator using the checklist and assist.
 - Emphasize the need to check GPS database currency.
 - For pertinent systems, show the student the GPS self-test (half-left, half-up, flag out of view).
 - Point out the RMI and OBS indications on that self-test.
 - The AHRS may take several minutes to align, depending on the system (the instructor may 'quick align').
 - Ensure that there are no red 'x's.
 - Have the student start the engine using the checklist.
- Instrument cockpit check:
 - Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.
 - Demonstrate to the student what needs to be done for each item.
 - Discipline the student to set bugs, bearing pointers, and any auxiliary features.
- Before taxi checks:
 - Student Tells, Student Does.
 - Explain to the student what the proper indications are for each instrument.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: discuss standby battery check and adjust the miniature aircraft (cannot be performed in the ATD).
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.

Instrument Rating Course

- Vertical speed indicator (PFD): reads 0 or use current indication.
- Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
- HSI: matches compass and turns appropriately.
- Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.
- Use a logical flow while checking instruments!
- Discuss the proper actions to take if a piece of equipment fails its check.

Give the student a departure via radar vectors. The instructor simulates ATC. Have the student switch to the appropriate frequencies throughout the simulator lesson.

Basic Instrument Maneuvers

- Straight and level:
 - Student Tells, Student Does.
 - Remind the student to begin level-off 10% vertical speed.
 - Assist the student with pitch and power settings for the simulator.
 - Allow the student to get a feel for trimming.
 - Training cruise for the appropriate aircraft is to be maintained unless otherwise specified.
 - The appropriate checklists must be completed.
- Pattern G:
 - Instructor Tells, Student Does.
 - This will account for entering constant airspeed climbs from cruise speed.
 - Pattern G will review the effect of pitch and power while entering climbs/descents from cruise airspeed.
 - Climbs are entered by reducing power before attitude and power are adjusted to maintain Vy.
 - Descents are entered by slowing to the proper speed before pitching down.
 - Monitor performance and coach use of pitch and power – ensure level off begins 10% VSI.
- Pattern H:
 - Instructor Tells, Student Does.
 - This will account for constant airspeed climbs from a specified speed.
 - Pattern H disciplines the student to coordinate pitch and power to maintain airspeed while climbing and descending.
 - Climbs & descents are entered by simultaneously using pitch and power.
 - Monitor performance and coach use of pitch and power – ensure level off begins 10% VSI.
- Vertical S:
 - Instructor Tells, Student Does.

- Specify a heading to begin the pattern.
- The instructor will call out the climbs, descents, and keeps time.
- Time and altitude will indicate the accuracy of flying – accurate flying is shown by reaching target altitudes at 1:00.
- Monitor the flying and coach use of pitch and power – level offs should begin 10% VSI.
- Once complete, note the time.
- Pattern A:
 - Instructor Tells, Student Does.
 - The instructor will call the turns and track time.
 - Ensure standard rate turns are established – coach the student on the approximate bank angle.
 - Monitor accuracy of turns and ensure rollouts begin ½ bank angle.
 - Upon completion of the maneuver, compare the student's time and the required time.
 - The original heading should be returned to.
- Pattern B:
 - Instructor Tells, Student Does.
 - The instructor will call the turns and track time.
 - The instructor will announce when to begin timing, and the student will keep track of the pattern.
 - Ensure standard rate turns are established – coach the student on the approximate bank angle.
 - Monitor accuracy of turns, timekeeping, and ensure rollouts begin ½ bank angle.
 - Upon completion of the maneuver, compare the student's time and the required time.
 - The original heading should be returned to.
- Pattern C:
 - Instructor Tells, Student Does.
 - The instructor will call the turns and track time.
 - Ensure standard rate turns are established – coach the student on the approximate bank angle.
 - Monitor accuracy of turns and ensure rollouts begin ½ bank angle.
 - Upon completion of the maneuver, compare the student's time and the required time.
 - The original heading should be returned to.
- Steep turns:
 - Student Does, Instructor Evaluates.
 - Note heading – conduct clearing turns at 30° bank preceding the maneuver.
 - Coach the student in holding the correct pitch and power setting.
 - Scan will need to be rapid to remain within parameters.
 - It is best to alternate directions for proficiency.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 303.
- Assign Homework:
 - Review the maneuvers' manual on unusual attitudes.
 - Review the Instrument Flying Handbook:
 - Chapter 07 - Section 21: timed turns and compass turns.
 - Chapter 07 - Section 26-28: unusual attitudes.

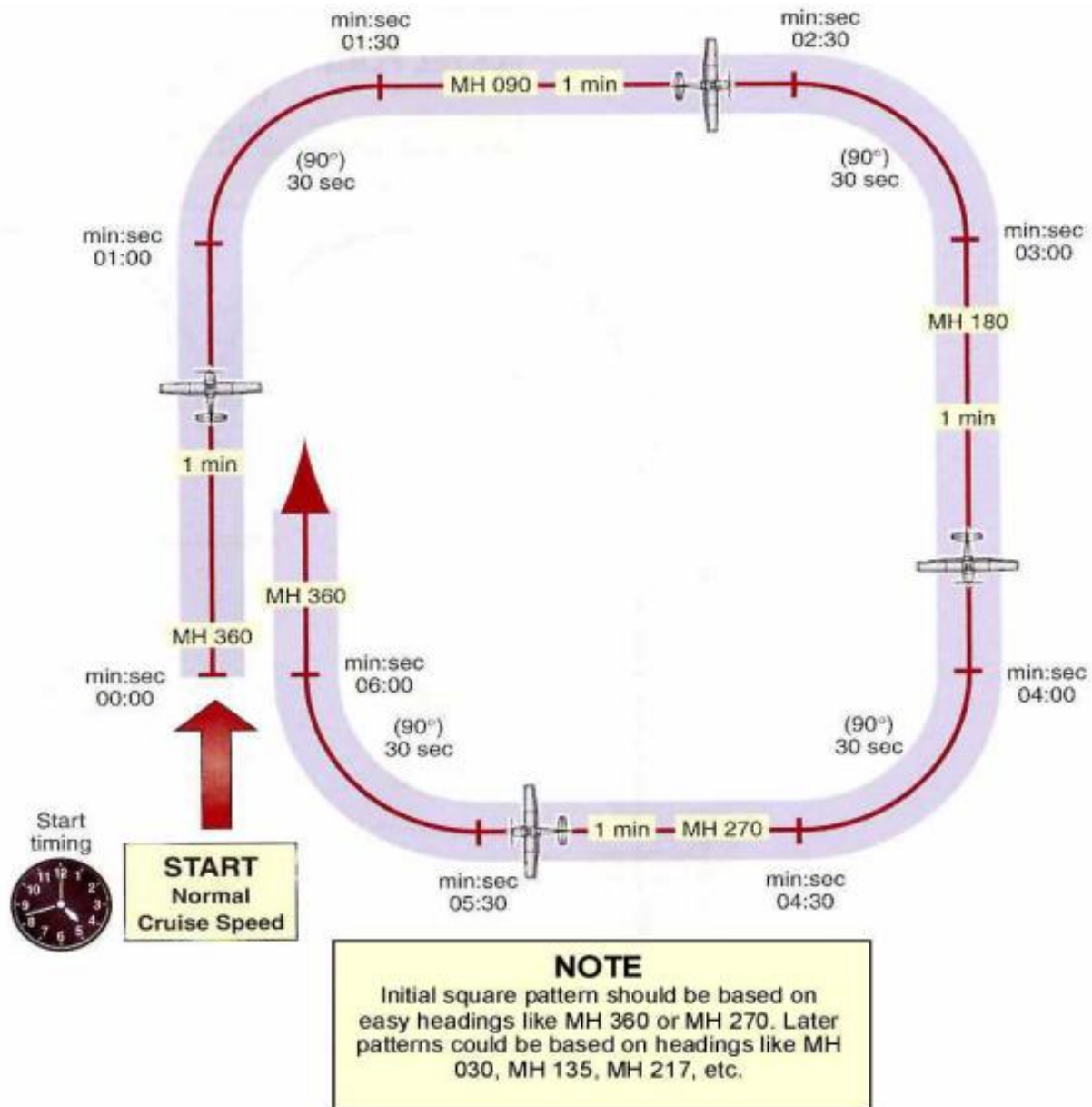
Completion Standards

This lesson will be complete when the student can, with instructor assistance, demonstrate aircraft control through the use of pitch, bank, and power during basic attitude instrument flight.

Lesson Plans

Instrument Rating Course

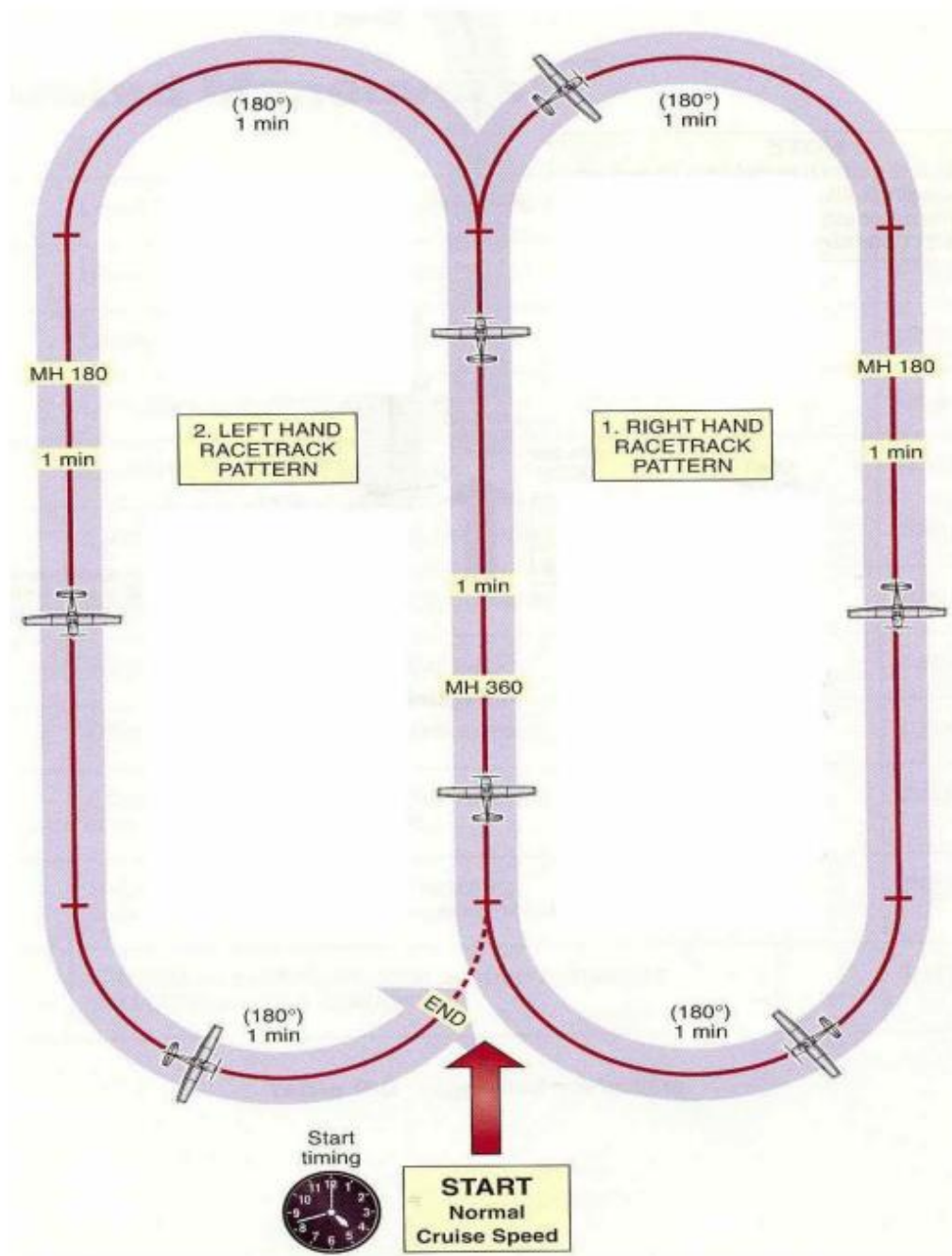
Pattern A



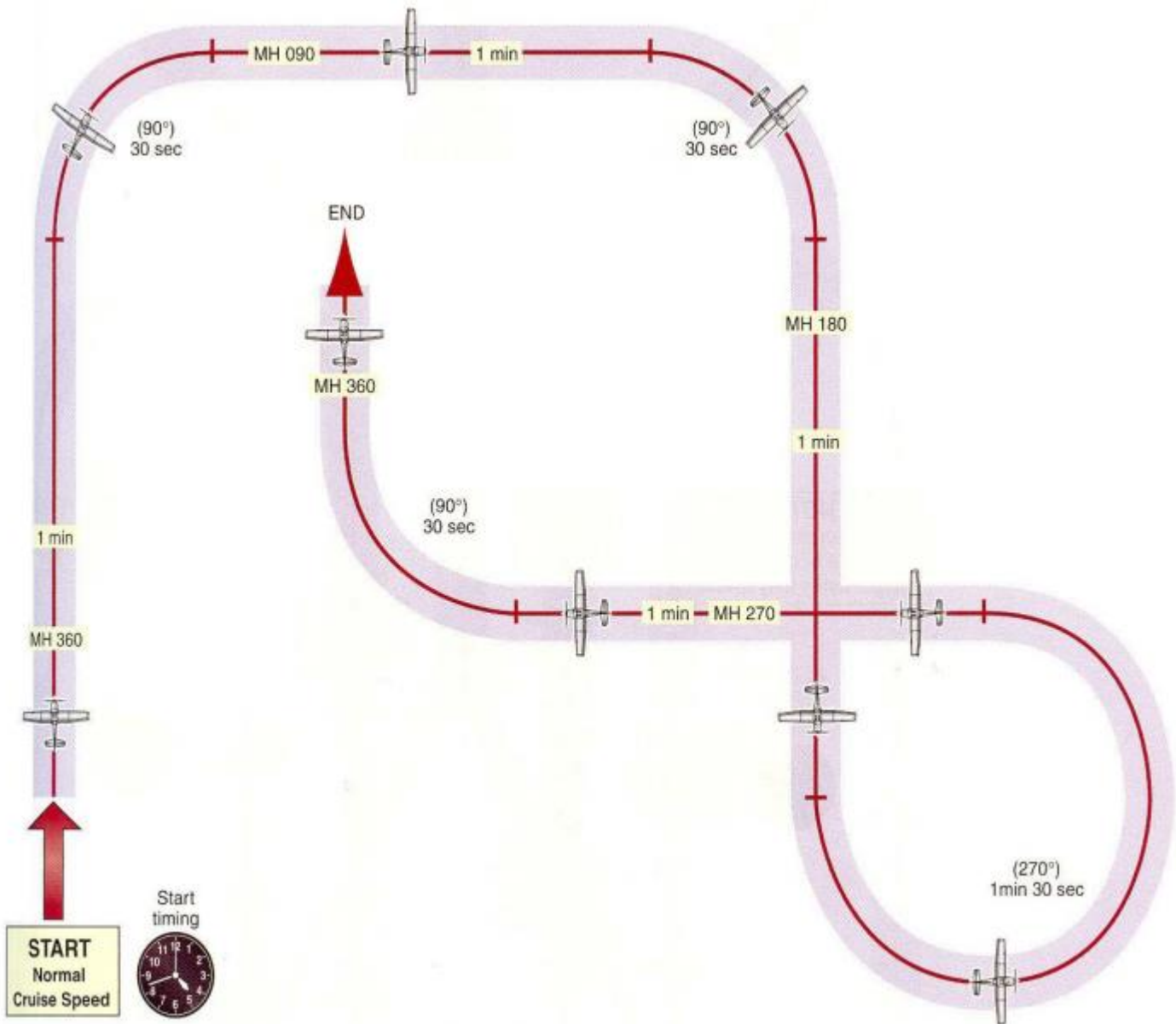
Lesson Plans

Instrument Rating Course

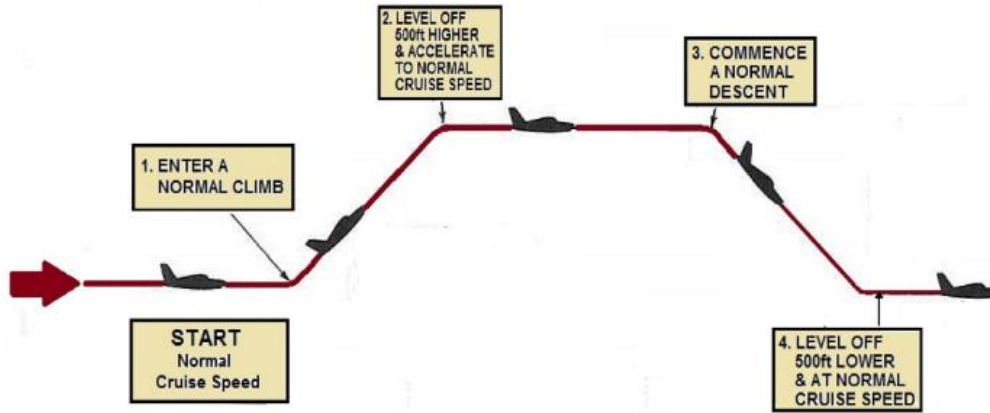
Pattern B



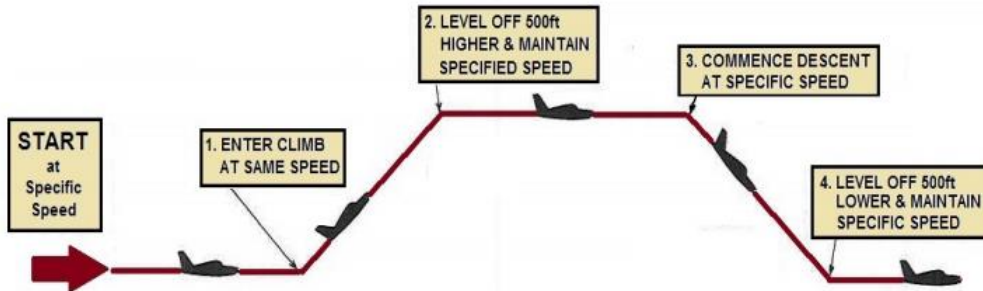
Pattern C



Pattern G



Pattern H



Vertical S

